ENME 351 Lab 8

Op-Amp Circuits with Arduino

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Objective

The purpose of this lab was to build and test several operational amplifier circuits using an Arduino and an LM741 op-amp. The circuits tested were a comparator, a unity gain voltage follower, an inverting amplifier, and an integrating amplifier. The Arduino measured the circuit voltages, while Processing displayed the real-time oscilloscope-style plots.

Arduino and Processing Setup

The Arduino measured four voltage signals:

Signal	Arduino Pin	Plot Color
V_{in}	A0	White
V_-	A1	Green
V_o	A2	Blue
V_+	A3	Yellow

The Arduino sampled the voltages every 50 ms and sent the values to Processing. Processing plotted each voltage against time using the required oscilloscope-style layout.

Comparator Circuit

Circuit Description

The comparator circuit used the LM741 in open-loop mode. A voltage divider made from two equal resistors created a fixed reference voltage of approximately 2.5 V. This reference was connected to the inverting input, while the potentiometer output was connected to the non-inverting input.

$$V_- = 2.5 \text{ V}$$

$$V_+ = V_{in}$$

When V_{in} was below 2.5 V, the output saturated low and the LED remained off. When V_{in} exceeded 2.5 V, the output saturated high and the LED turned on.

$$V_{in} < 2.5 \text{ V} \Rightarrow V_o \approx 0 \text{ V}$$

$$V_{in} > 2.5 \text{ V} \Rightarrow V_o \approx 5 \text{ V}$$

Comparator Circuit Photos

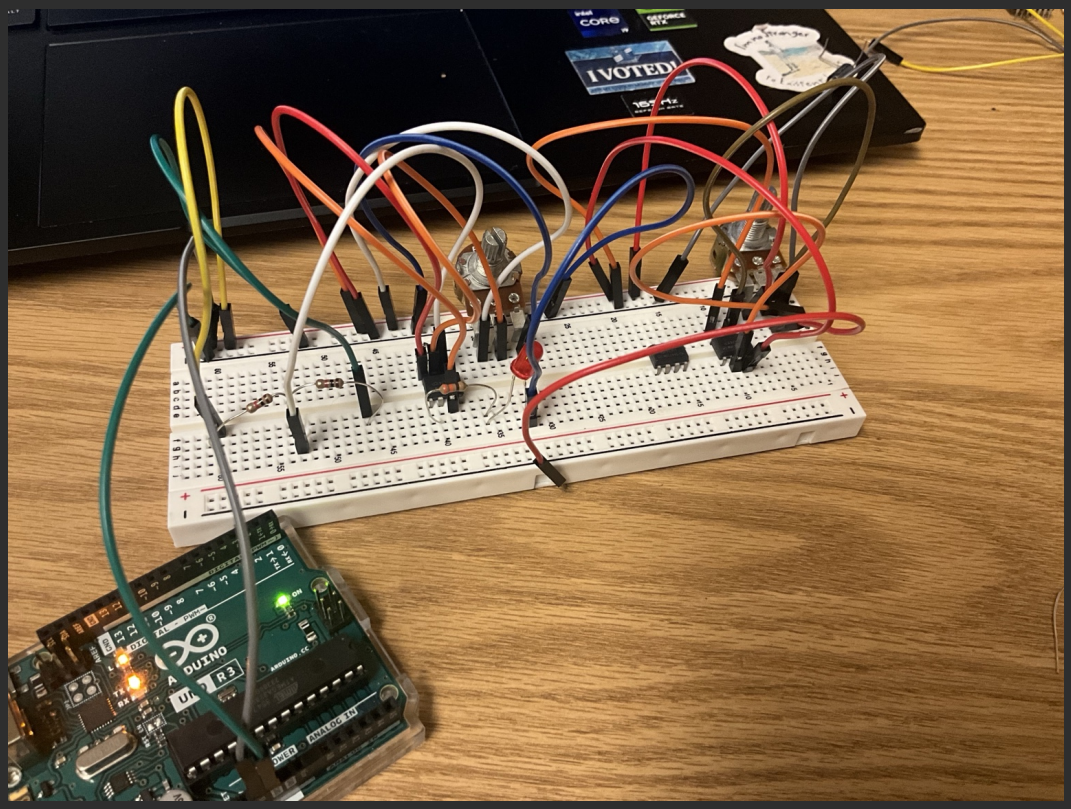

Figure 1: Comparator circuit with LED off. The potentiometer input is below the 2.5 V threshold.

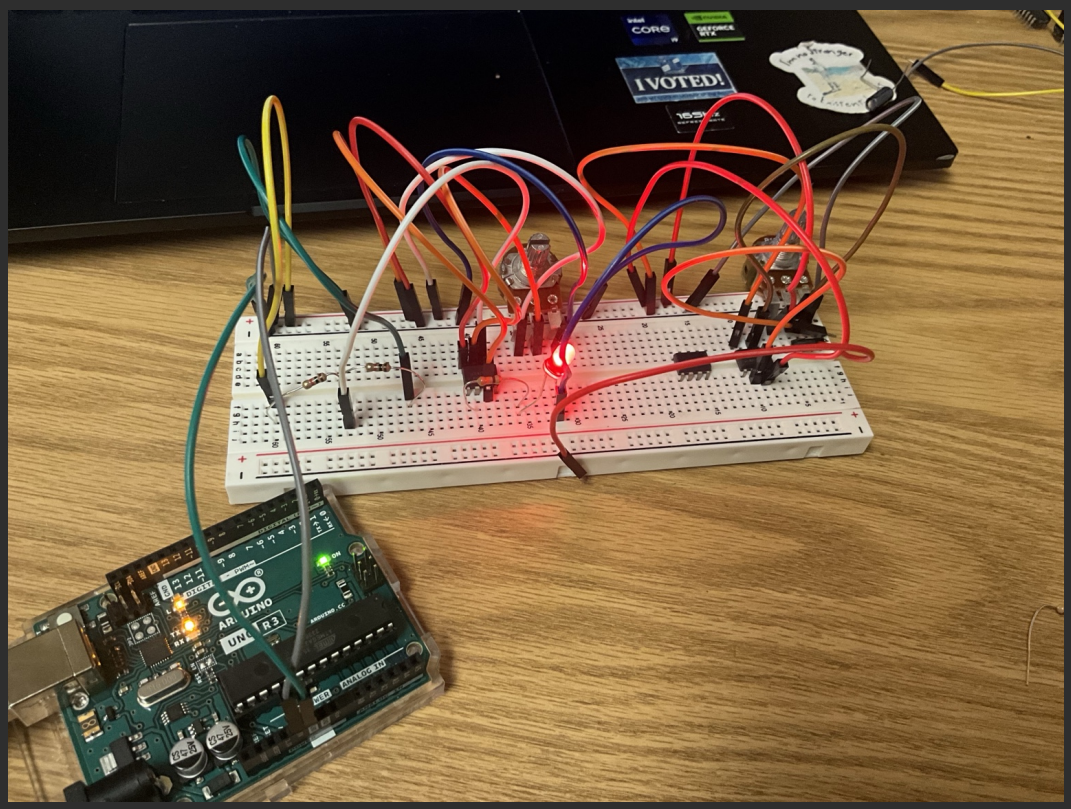

Figure 2: Comparator circuit with LED on. The potentiometer input is above the 2.5 V threshold.

The comparator behaved like a switch rather than a smooth amplifier. Once the potentiometer voltage crossed the threshold, the LED turned on. This makes sense because the LM741 was used open-loop, so a small difference between V_+ and V_- caused the output to saturate.

Exercise 1: Non-Inverting Amplifier Analysis

Exercise 1 required deriving the gain of the non-inverting amplifier.

For a non-inverting amplifier:

$$V_+ = V_{in}$$

With negative feedback, the ideal op-amp rule gives:

$$V_- \approx V_+$$

Therefore:

$$V_- = V_{in}$$

The inverting input is connected to a voltage divider between V_o and ground:

$$V_- = V_o \frac{R_1}{R_1 + R_2}$$

Substitute $V_- = V_{in}$:

$$V_{in} = V_o \frac{R_1}{R_1 + R_2}$$

Solving for V_o:

$$V_o = V_{in} \frac{R_1 + R_2}{R_1}$$

$$V_o = V_{in} \left(1 + \frac{R_2}{R_1}\right)$$

Thus, the closed-loop gain is:

$$\boxed{G = \frac{V_o}{V_{in}} = 1 + \frac{R_2}{R_1}}$$

Exercise 2: Unity Gain Amplifier / Voltage Follower

Circuit Description

The voltage follower connects the output directly back to the inverting input:

$$V_- = V_o$$

The potentiometer output is connected to the non-inverting input:

$$V_+ = V_{in}$$

Since feedback forces $V_- \approx V_+$:

$$V_o \approx V_{in}$$

Therefore, the gain is:

$$\boxed{G = 1}$$

Circuit Photo

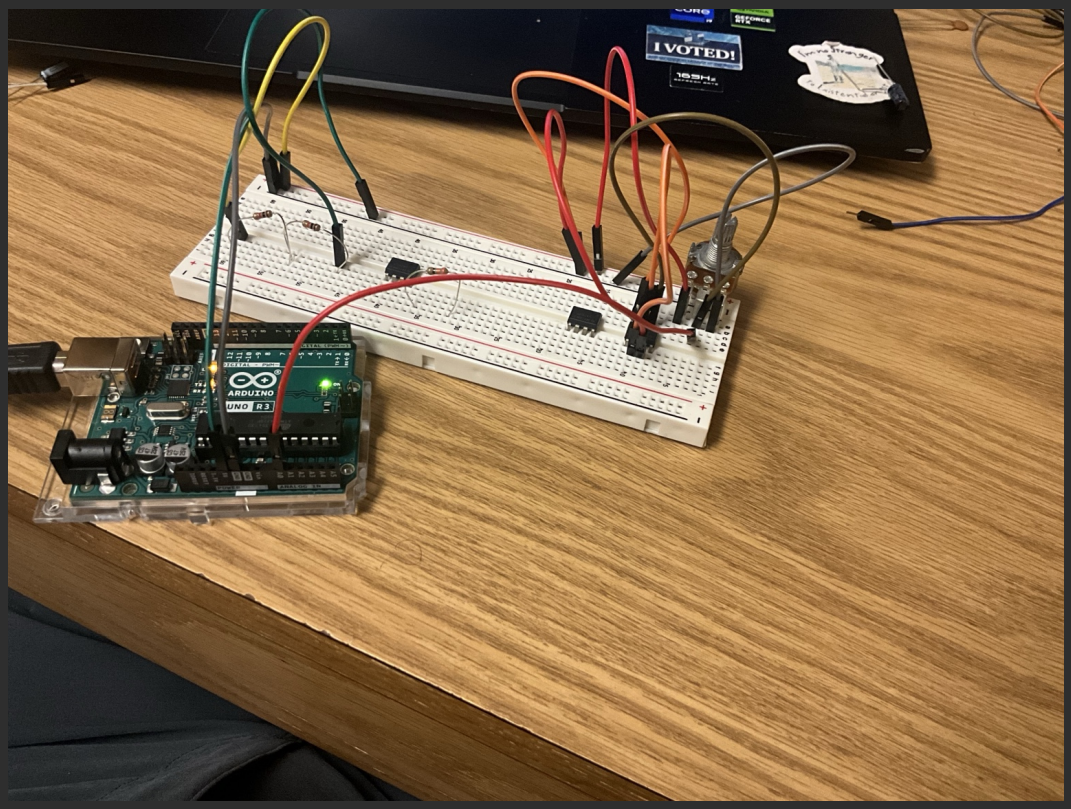

Figure 3: Exercise 2 voltage follower circuit. The output is directly fed back to the inverting input.

Oscilloscope Plot

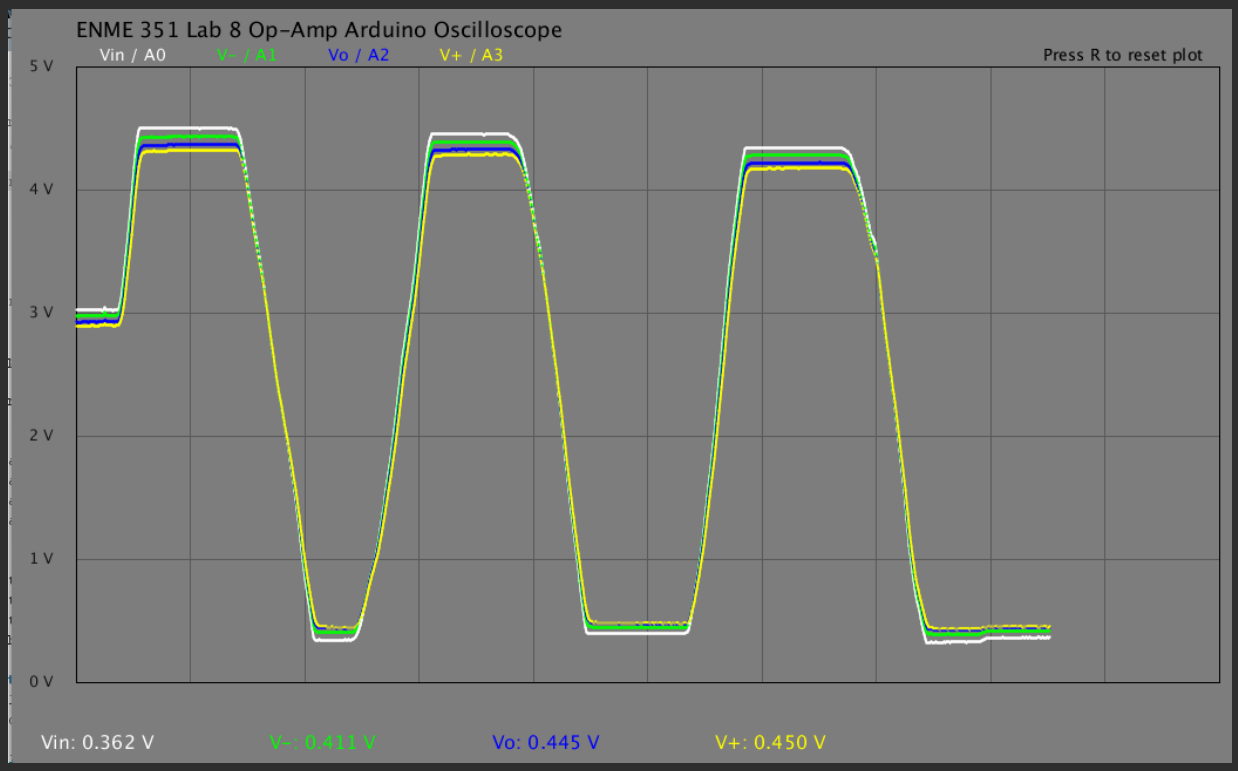

Figure 4: Voltage follower plot showing V_o tracking V_{in}.

The voltage follower produced approximately unity gain. The white input trace, green inverting input trace, blue output trace, and yellow reference trace followed closely. This confirms that the output was approximately equal to the input. Small differences are expected because the LM741 is not ideal and is not rail-to-rail.

Exercise 3: Inverting Amplifier

Circuit Description

The inverting amplifier used:

$$R_1 = 1k\Omega$$

$$R_2 = 10k\Omega$$

The non-inverting input was biased at 2.5 V because the Arduino only provides a 0–5 V supply. Therefore, 2.5 V acted as a virtual ground.

The gain is:

$$G = -\frac{R_2}{R_1}$$

$$G = -\frac{10k\Omega}{1k\Omega}$$

$$\boxed{G = -10}$$

Since the circuit is centered around 2.5 V:

$$\boxed{V_o - 2.5 = -10(V_{in} - 2.5)}$$

This means that when V_{in} rises above 2.5 V, V_o decreases. When V_{in} falls below 2.5 V, V_o increases.

Circuit Photo

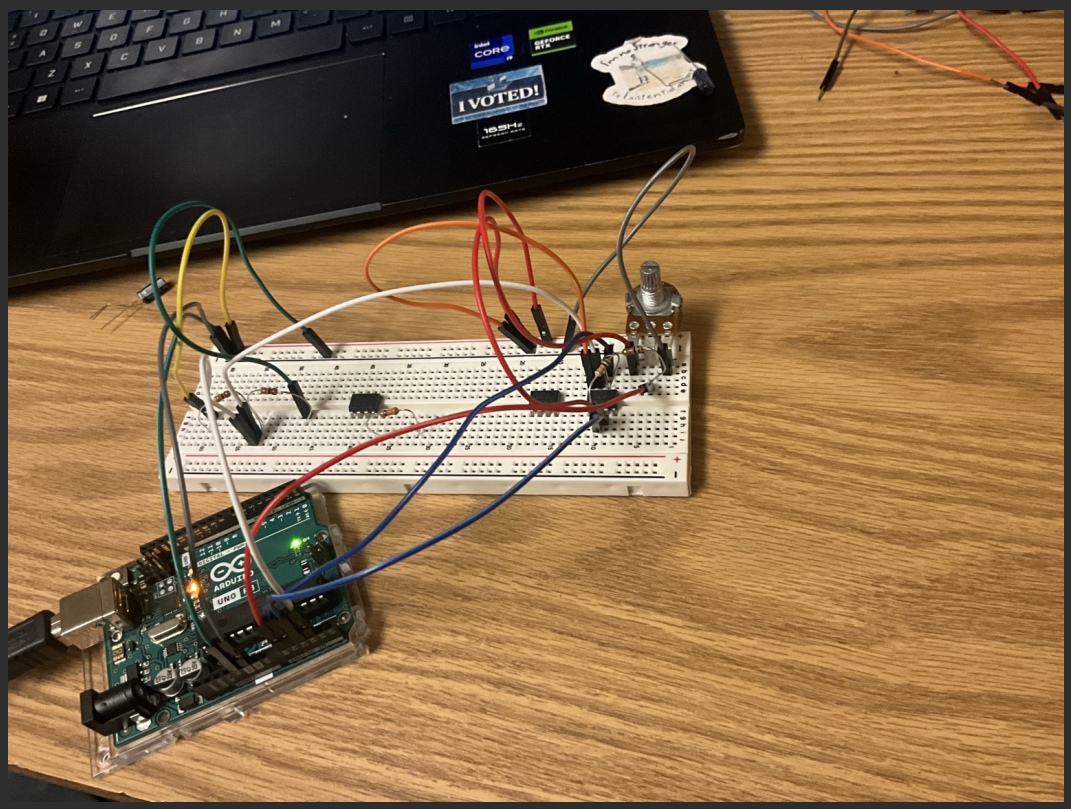

Figure 5: Exercise 3 inverting amplifier circuit using $R_1 = 1k\Omega$ and $R_2 = 10k\Omega$.

Oscilloscope Plot

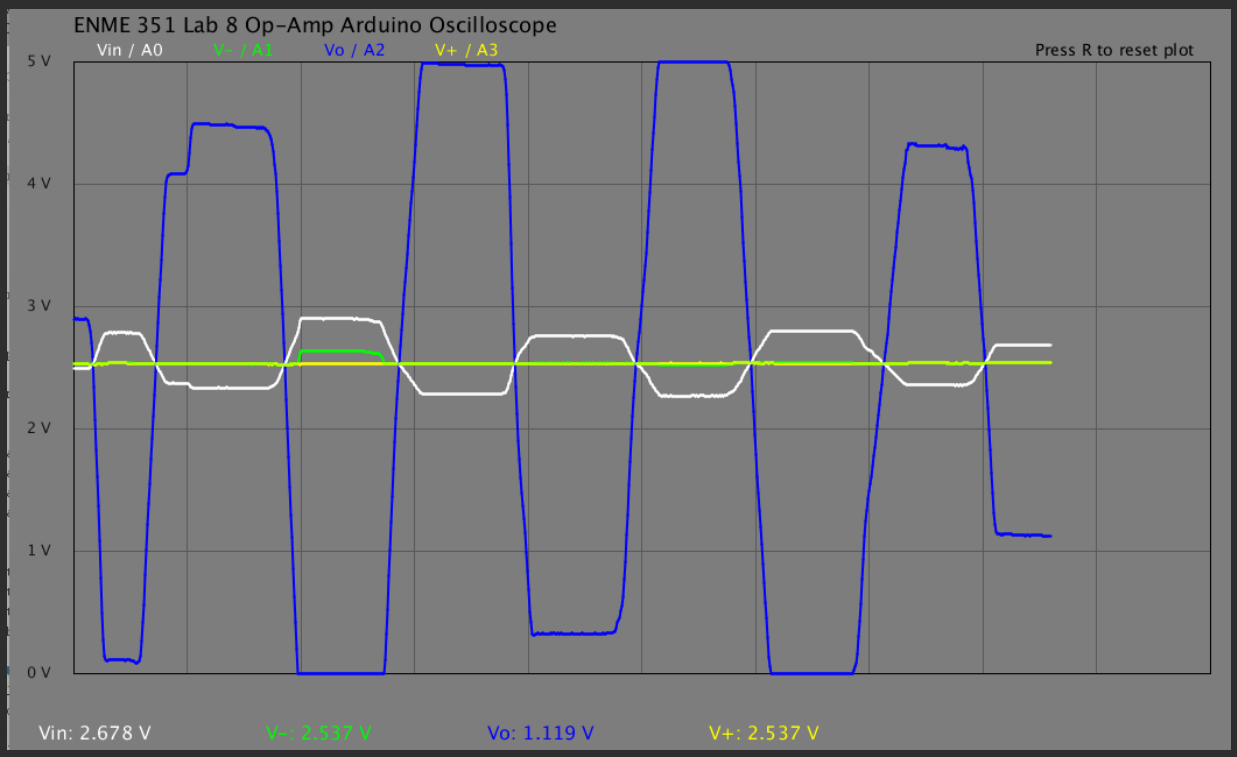

Figure 6: Inverting amplifier plot showing output inversion and saturation.

The output moved opposite the input, confirming inverting behavior. Because the gain magnitude was 10, small changes around 2.5 V produced large output changes. The blue output trace saturated near 0 V and 5 V because the Arduino supply limited the op-amp output range.

Exercise 4: Integrating Amplifier

Circuit Description

The integrator circuit replaced the feedback resistor with a capacitor. The input resistor was:

$$R = 4.7k\Omega$$

and the feedback capacitor was:

$$C = 10\mu F$$

The non-inverting input was again biased at 2.5 V. Thus, the circuit integrated the input relative to 2.5 V.

The current through the input resistor is:

$$i_R = \frac{V_{in} - 2.5}{R}$$

For the capacitor:

$$i_C = -C\frac{dV_o}{dt}$$

Setting $i_R = i_C$:

$$\frac{V_{in} - 2.5}{R} = -C\frac{dV_o}{dt}$$

Therefore:

$$\frac{dV_o}{dt} = -\frac{V_{in} - 2.5}{RC}$$

Integrating both sides gives:

$$\boxed{V_o(t) = V_o(0) - \frac{1}{RC}\int_0^t (V_{in}(\tau) - 2.5)\, d\tau}$$

With:

$$R = 4.7k\Omega$$

$$C = 10\mu F$$

$$RC = (4700)(10 \times 10^{-6}) = 0.047 \text{ s}$$

Circuit Photo

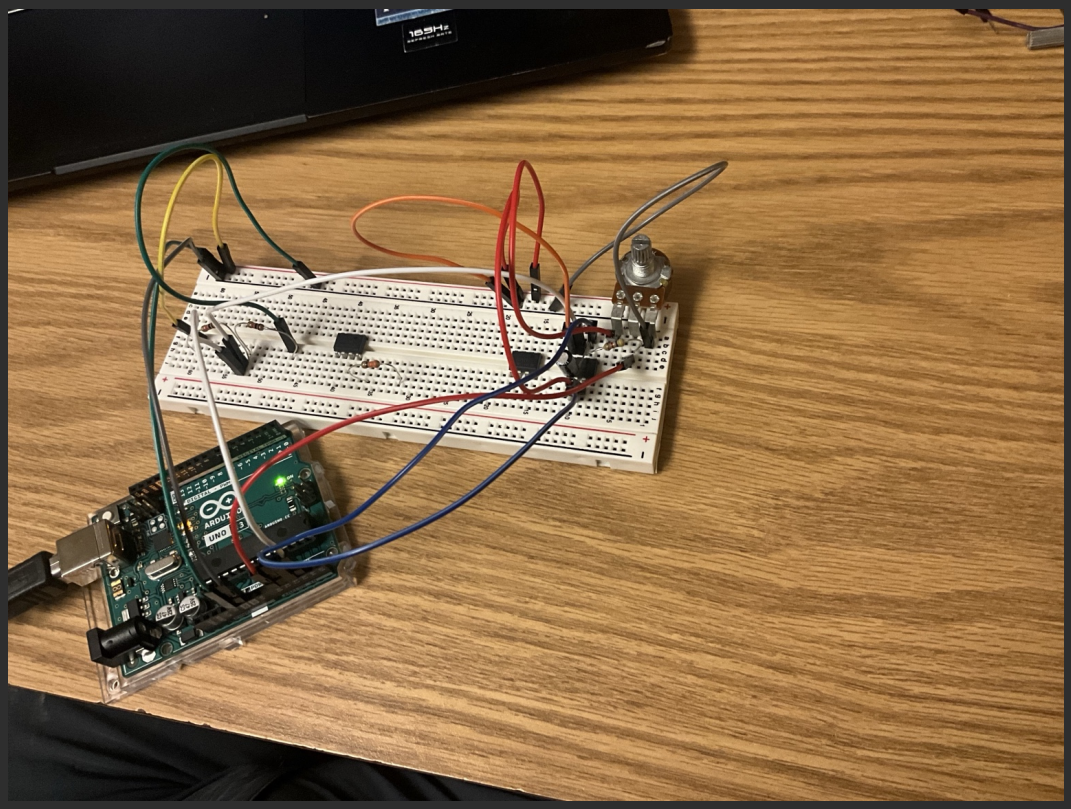

Figure 7: Exercise 4 integrating amplifier circuit using a 4.7 kΩ input resistor and 10 μF feedback capacitor.

Oscilloscope Plot

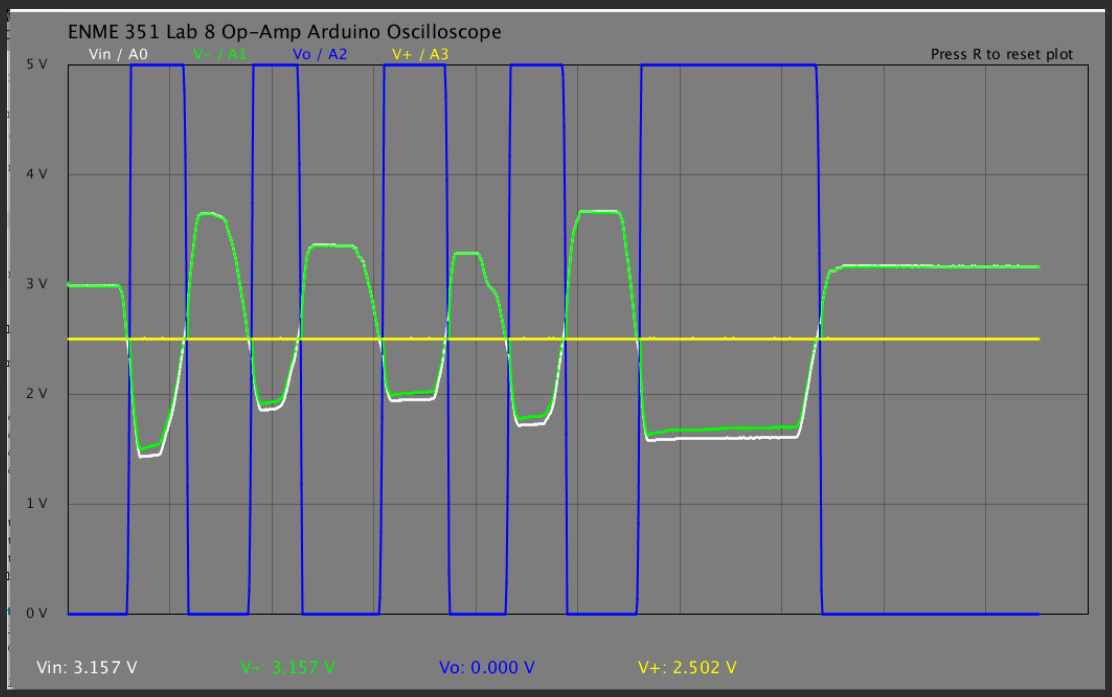

Figure 8: Integrator plot showing ramping behavior and output saturation.

The integrator output changed over time instead of responding immediately. When V_{in} was above 2.5 V, the output ramped downward. When V_{in} was below 2.5 V, the output ramped upward. Once the output reached the rail limits, it saturated near 0 V or 5 V.

Comparison of Circuits

Circuit	Main Component / Feedback	Behavior
Comparator	No feedback	Switches LED on/off at threshold
Voltage Follower	Direct wire feedback	$V_o \approx V_{in}$
Inverting Amplifier	10 kΩ feedback resistor	Output is inverted and amplified
Integrator	10 μF feedback capacitor	Output ramps over time

Conclusion

This lab demonstrated several important op-amp configurations. The comparator showed how an op-amp can be used as a threshold detector. The voltage follower demonstrated unity gain buffering, where the output closely followed the input. The inverting amplifier showed how resistor feedback controls gain and inversion. Finally, the integrator showed how replacing the feedback resistor with a capacitor makes the output depend on the accumulated input over time.

Overall, the results matched the expected behavior. The main practical limitation observed was saturation near the 0 V and 5 V rails, which occurred because the LM741 was operated using a single Arduino supply.